inzva DLSG Assignment 2
Structural Inductive Bias in Graph Neural Networks

Name Surname,

This report must follow a research-style format. The objective is to conduct a controlled empirical study analysing how structural inductive bias affects graph classification performance.

1 Introduction

In this project, you will study the role of *structural inductive bias* in graph neural networks (GNNs).

You must: Formally define graph classification. Explain message passing.

2 Problem Analysis

You will evaluate the following datasets:

OGB Molecular Benchmarks: *ogbg-molhiv*

TU Graph Benchmarks: *proteins*, *imdb-multi*, *reddit-binary*

For each dataset, analyse: Number of graphs and average graph size, node feature dimensionality, task type (binary / multi-class / multi-label), class imbalance, whether edges are semantically meaningful

3 Method

You must compare the following models:

Feature-Only Models LinearBaseline (Mean pooling + Logistic Regression), MLPBaseline (Mean pooling + MLP), DeepSets

Message Passing Models GCN, GraphSAGE, GAT, GIN

You must clearly report:

- Hidden dimension
- Number of layers
- Aggregation and readout functions
- Activation functions
- Regularisation strategy
- Total parameter count

4 Experiments

Your experimental protocol must ensure fairness and reproducibility: At least 3 random seeds with same parameter setups. Use Official OGB splits Report mean $\pm$ standard deviation for the experimentation results. (That's why you use seeds for the same setup.)

Use: ROC-AUC for OGB datasets and just accuracy for TU datasets

This is how you cite [1]. Cite the models and datasets you have used.

Table 2: Model complexity comparison.

Model	Parameter Count
LinearBaseline	
MLPBaseline	
DeepSets	
GCN	
GraphSAGE	
GATv2	
GIN	

Table 1: Performance comparison across models. Report mean ± std. Best results in **bold**.

Model	ogbg-molhiv	PROTEINS	IMDB-MULTI	REDDIT-BINARY
LinearBaseline				
MLPBaseline				
DeepSets				
GCN				
GraphSAGE				
GAT				
GIN				

Figure 1: Example visualisation: performance comparison or depth vs accuracy analysis. You are free to be creative.

5 Results

In this section, you must: Compare feature-only and message passing models, identify dataset-dependent trends, analyse whether higher expressivity improves performance, in other words, answer does increasing the model complexity provides better accuracy? Discuss evidence of overfitting or oversmoothing.

6 Conclusions

Summarise:

- Main empirical findings
- Limitations of the study (GPU power :D)
- Suggestions for future work (Trying experiments with more GPU power:D)

References

[1] Y. LeCun, Y. Bengio, and G. Hinton. Deep learning. *nature*, 521(7553):436–444, 2015.